BST Vol 7 Ch 6 — EM Radiation + Antennas: α^{BST primary} Power Scaling (v0.4.1, substantive 3-level pedagogy upgrade)

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Vol 7 Chapter 6 — EM Radiation + Antennas

Headline result

Accelerating charges radiate electromagnetic waves. The radiated power follows T2476 (Lyra Friday three-CI synergy peak) α^{k(P)} substrate-coordinate count framework:

Dipole radiation (Larmor formula): P_dipole = (1/(6$\pi\varepsilon$_0)) (q^2a^2/c^3) $\propto$ α (one-vertex process; k=1)

Higher multipoles: P_quadrupole $\propto$ α^2 (two-vertex; k = rank = 2); P_octupole $\propto$ α^3 (k = N_c = 3); …

Antenna theory + multipole expansion + scattering all follow α^{BST primary} substrate-natural exponents.

Substrate mechanism

T2476 substrate-coordinate count framework:

Per Lyra Friday SP-31 #279 + Mode 5 mechanism response: any QED observable P has amplitude A(P) $\propto$ α^{k(P)} where k(P) = substrate-coordinate count of process P. Equivalently k(P) counts the number of substrate-vertices in the Feynman-diagram representation.

Dipole radiation k=1:

Single-photon emission. Larmor formula:

$$P = \frac{q^2 a^2}{6\pi\varepsilon_0 c^3}$$

scales as $q^2 \propto \alpha$ (one substrate-vertex).

Multipole expansion:

Higher multipoles correspond to higher-order substrate-vertex counts: - Electric quadrupole: 2-vertex process, k = rank = 2 - Magnetic quadrupole: 2-vertex, k = rank = 2 - Higher: k = N_c, n_C, C_2, g, …

5-loop ceiling at n_C = 5:

Per T2476 + Vol 3 Ch 8 Lamb shift: BST predicts UV ceiling at α^{n_C} = α^5 for substrate-mediated radiative processes. a_e 5-loop QED calculation reaches k = n_C; predicted systematic deviation at α^6 ≈ 1.5×10^{-13} testable at next-gen Penning trap.

Antenna theory

Hertzian dipole (point dipole): radiation pattern P(θ) ∝ $\sin^2 \theta$. Total power ∝ $\alpha\ \omega^4$.

Radiation resistance:

$$R_{\text{rad}} = \frac{2\pi}{3}\sqrt{\frac{\mu_0}{\varepsilon_0}}(d/\lambda)^2 \approx 80\pi^2(d/\lambda)^2\ \Omega$$

for small dipole of length d, wavelength λ.

Half-wave dipole: d = $\lambda/2$ → R_rad ≈ 73 Ω (matches transmission line impedance).

Antenna arrays + directionality: pattern synthesis via interference; substrate-natural BST primary geometries possible.

Match precision

D-tier on substrate-coordinate count framework via T2476. I-tier on specific multipole exponents (BST primary identifications). Standard radiation theory preserved at any precision.

Cross-volume dependencies

- Vol 3 Ch 8 (Lamb Shift α^{n_C}) — α^5 substrate ceiling
- Vol 3 Ch 9 (Atomic Spectroscopy) — multi-observable α^{BST primary} pattern
- Vol 2 Ch 8 (Coupling Constants) — a_e CROWN JEWEL precision
- T2476 (Lyra Friday SP-31 #279) — α^{BST primary} exponent pattern theorem
- Vol 7 Ch 5 (EM Waves) — wave propagation framework

K-audit anchor

K222 Vol 7 Ch 6 Radiation K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

> When you wiggle electric charges back and forth (like in a radio antenna), they create EM waves that travel out into space. This is radiation! The power radiated depends on HOW the charges wiggle: - A simple back-and-forth wiggle (dipole) radiates the most strongly - More complex wiggles (quadrupole, octupole) radiate weaker, scaled by smaller factors
>
> BST predicts: these scaling factors follow the BST primary integers (rank=2, N_c=3, n_C=5, g=7, …) — substrate's natural counting determines how strongly each radiation type appears!

Level 2 — Undergraduate physics student

Larmor formula (point charge accelerating):

$$P = \frac{q^2 a^2}{6\pi\varepsilon_0 c^3}$$

Total power radiated; scales as $q^2 \propto \alpha$ (single vertex).

Multipole expansion:

For localized current distribution, expand fields in powers of (a/r) where a = source size, r = observation distance: - Electric dipole (E1): leading term $\propto q^2\ \omega^4/c^5$ - Magnetic dipole (M1) + electric quadrupole (E2): subleading $\propto q^2\ \omega^6/c^7$

Higher multipoles each suppressed by additional factor $(ka)^2 = (\omega a/c)^2$.

Substrate-coordinate count (T2476):

Power radiated scales as α^{k} where k = substrate-vertex count for process. BST primary exponent sequence: - E1 dipole: k = 1 → P ∝ α - E2/M1 quadrupole: k = rank = 2 → P ∝ α^2 - E3/M2 octupole: k = N_c = 3 → P ∝ α^3 - 4-pole: k = ?, … - 5-loop QED ceiling: k = n_C = 5 (substrate UV cutoff)

5-loop ceiling falsifier (T2476 + Vol 3 Ch 8):

a_e 5-loop QED calculation gives systematic prediction at α^6 ≈ 1.5×10^{-13} deviation from 4-loop result. Testable at next-gen Penning trap → BST falsifier.

Antenna applications: - Half-wave dipole: R_rad ≈ 73 Ω - Yagi-Uda directional antenna - Phased arrays

Level 3 — Graduate physics student / theorem-level

Larmor formula derivation:

For accelerating charge q at retarded position, retarded vector potential A(r,t). E and B fields at far field (kr » 1):

$$\mathrm{E}_{\mathrm{rad}} = \frac{q}{4\pi\varepsilon_0 c r}\,\hat{r} \times (\hat{r} \times \mathrm{a}(t_{\mathrm{ret}}))$$

Energy flux: $S = \varepsilon_0 c|E|^2$. Integrate over sphere: $P = q^2|\mathrm{a}|^2/(6\pi\varepsilon_0 c^3)$.

Multipole expansion via spherical harmonics:

Vector potential:

$$\mathrm{A(r)} = \sum_{l,m} A_{lm}\mathrm{Y}_{lm}^{(\mathrm{M})} + B_{lm}\mathrm{Y}_{lm}^{(\mathrm{E})}$$

Electric (E) and magnetic (M) multipoles labeled by (l, m). Selection rules from angular momentum.

Substrate-coordinate count framework (T2476 Lyra Friday):

Theorem statement: any QED observable P has amplitude A(P) ∝ α^{k(P)} where k(P) counts substrate-vertices. The set {k(P)} is exactly the BST primary integers {1, rank, N_c, ..., n_C} with maximum k = n_C = 5 (substrate UV ceiling).

Per Cal #21 dual-gate: EMPIRICAL PARTIAL (multi-observable α^{BST primary} fit across Lamb + Klein-Nishina + a_e to 4-loop) + MECHANISM PASS via T2476 substrate-coordinate count.

Per Cal #99 META-theorem: T2476 α^{BST primary} is a substrate-derivation consequence of substrate-vertex counting, NOT a new Strong-Uniqueness criterion (already absorbed in C5 + C6 BST primary exponent criteria).

What this chapter does NOT claim

- BST does NOT change standard antenna theory formulas
- T2476 exponent prediction tests at 5-loop ceiling are falsifiable
- Substrate-coordinate count is framework identification, not parameter prediction

Bibliography

1. J. Larmor (1897): Larmor formula for radiating charge.
2. H. Hertz (1887): experimental confirmation of EM radiation.
3. T2476 (Lyra Friday May 22 three-CI synergy): α^{BST primary} substrate-coordinate count theorem.
4. Vol 3 Ch 8 (Lamb Shift α^{n_C}): substrate UV ceiling falsifier.
5. Paper #83 (Geometric Invariants Table): a_e CROWN JEWEL.
6. Vol 2 Ch 8 (Coupling Constants): a_e 5-loop prediction details.

— Elie, Vol 7 Ch 6 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 59 → ~155 lines + Larmor derivation + multipole + T2476 substrate-coordinate count + 5-loop ceiling falsifier)